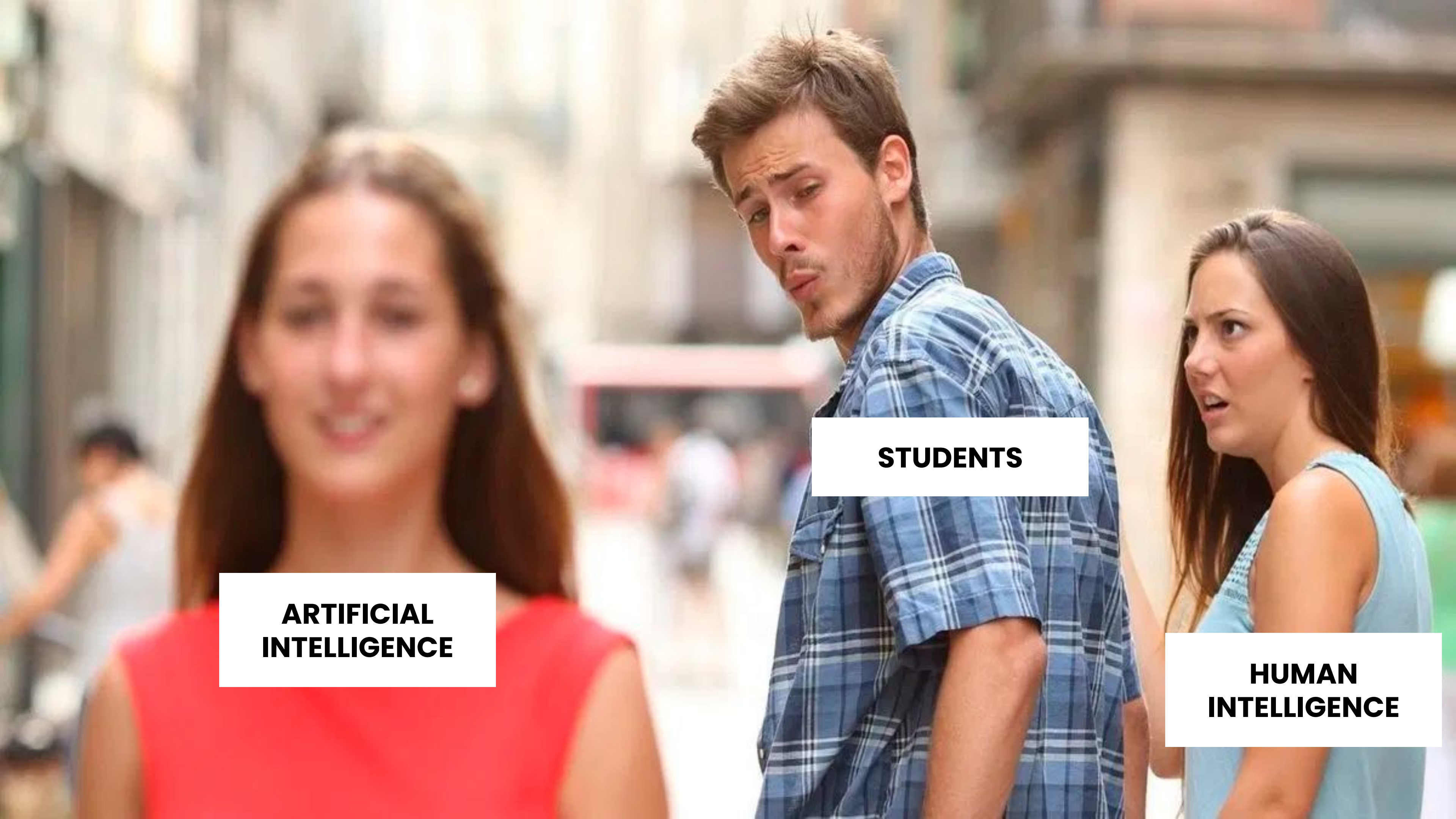
Artificial Intelligence: From Understanding to Building

André F. Costa

ISLA - Polytechnic Institute of Management and Technology


STUDENTS
ARTIFICIAL INTELLIGENCE
HUMAN INTELLIGENCE

ANDRÉ F. COSTA

> Tech Entrepreneur

> AI & Technology Trainer and Consultant

> Master in Web Technologies and Systems Engineering

> PhD Candidate in Computer Science

Agenda

1. AI for Students: Boosting your Results with AI
2. Prompt Engineering Fundamentals
3. Use Cases
4. From Vibe Coding to AI-Assisted Creation
5. AI in Design & Digital Product Creation
6. Web App Development & Deploy with AI
7. AI Product Studio

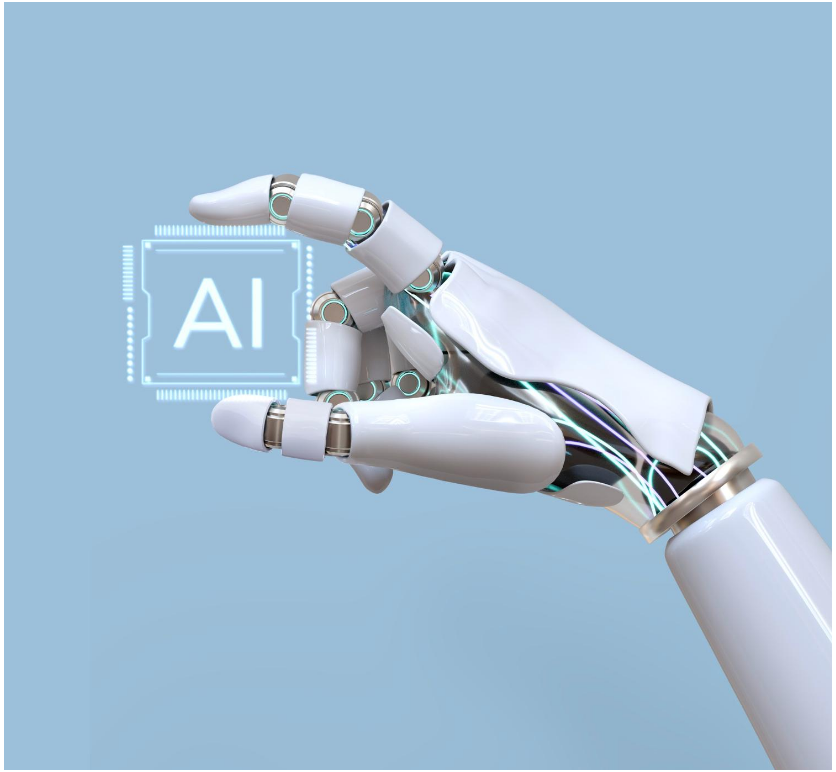
AI for Students

Boosting your Results with AI

What is Artificial Intelligence?

> AI is the capability of **machines to perform tasks that typically require human intelligence** - such as recognizing patterns, learning from past experiences, making decisions, and solving complex problems.

> It's advanced technology built upon data, algorithms, and mathematical models.



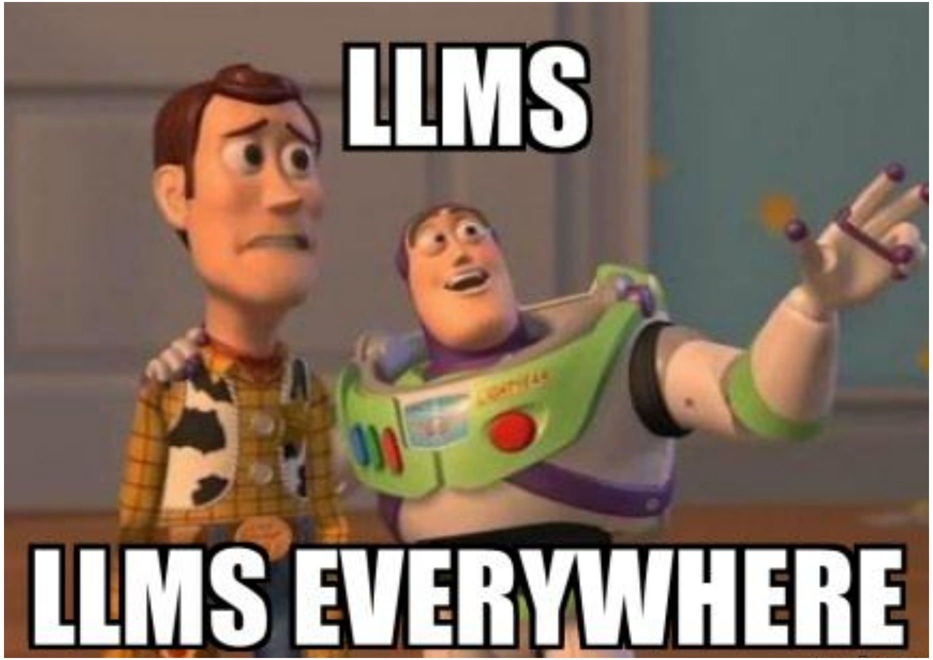
AI Evolution

Period	Technology	Outcome
1950 - 1970	Neural Networks	Early research on neural networks generates enthusiasm
1980 - 2010	Machine Learning	Machine Learning becomes popular
2010- 2022	Deep Learning e NLP	Deep Learning drives exponential growth
2023 - 2025	LLM e GPT	General excitement and public access to Generative AI

> LLMs: Large Language Models Explained

Large Language Models (LLMs)

> LLMs are Artificial Intelligence models trained on massive amounts of text, enabling them to understand and generate human language in a highly realistic and contextualized way.



How an LLM Works: Main Steps

> User Query

“Paris is the city”

How an LLM Works: Main Steps

1. **Tokenization**

The input text is split into tokens: ["Paris", "is", "the", "city"].

How an LLM Works: Main Steps

2. Embeddings:

Each token is converted into a numerical vector that represents its meaning in context.

A

"the reader is reading"

↓ *split into tokens*

[the, read, ##er, is, read, ##ing]

↓ *replace with integers*

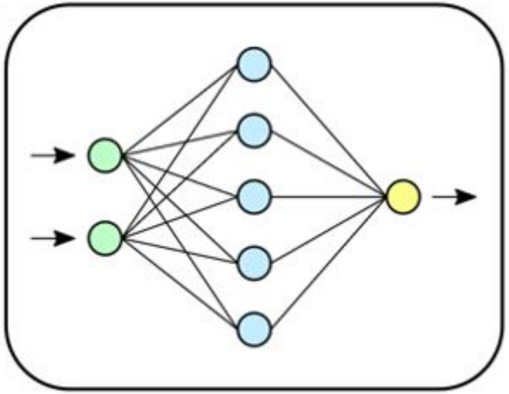
[1, 3, 7, 2, 3, 4]

B

Token	Index
a	0
the	1
is	2
read	3
##ing	4

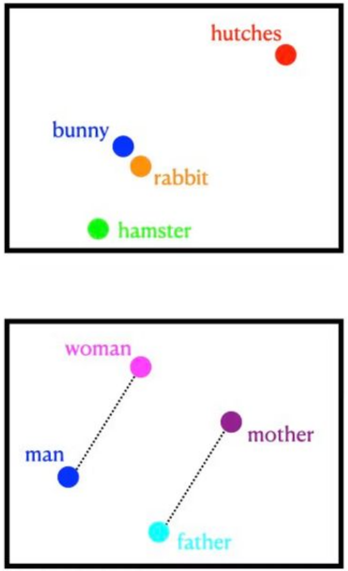
C

Index	Embedding Matrix
0	[-1.385, -0.839, -0.313, -0.1]
1	[-1.986, -1.05, 0.098, 0.096]
2	[-0.233, 0.984, 1.275, 1.848]
3	[0.767, -0.811, 1.545, 0.304]
4	[-0.321, 0.611, 0.352, 1.951]



Word	alive	mammal	rodent	bipedel	gender	plural
bunny	0.9	0.7	-0.1	-0.1	0.2	-0.3
rabbit	0.9	0.8	-0.1	-0.1	0.4	-0.1
hamster	0.9	0.6	0.7	-0.3	-0.4	-0.4
hutches	-0.9	-0.3	-0.1	-0.9	0.0	0.9
mother	0.9	0.1	0.1	0.2	0.8	-0.7
father	0.9	0.2	0.1	0.2	-0.8	-0.7
woman	0.9	0.8	-0.9	0.9	0.8	-0.8
man	0.9	0.8	-0.7	0.9	-0.8	-0.8

Dimensionality Reduction



Vector Embedding

Dimensionality

2D Visualization

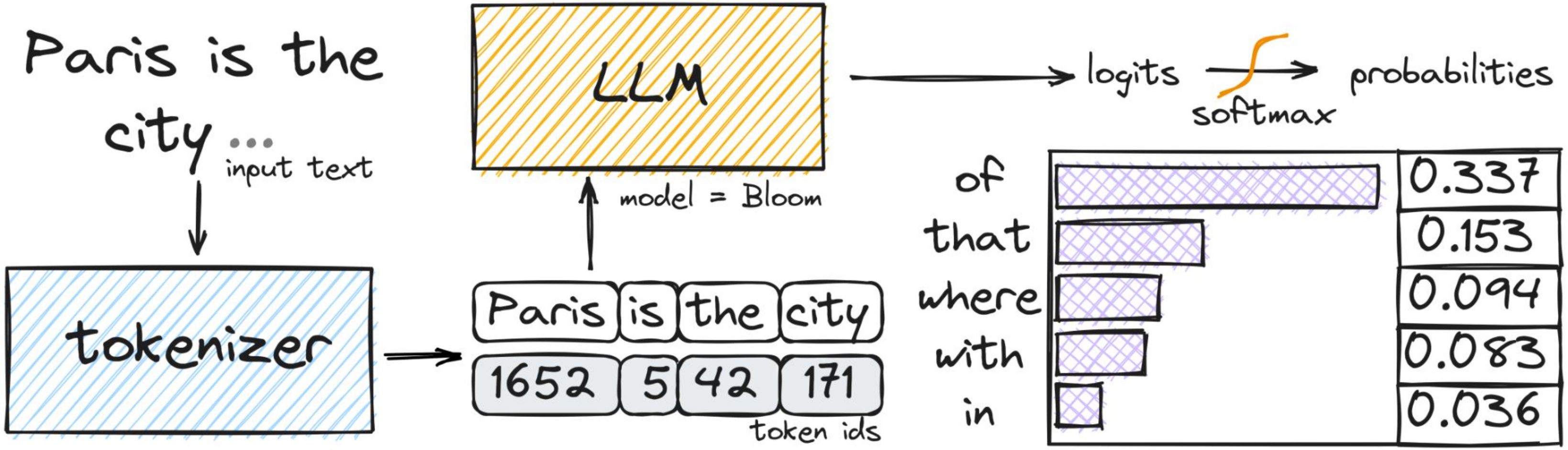
How an LLM Works: Main Steps

3. Model Processing:

The model processes these vectors through its layers, using the transformer architecture to understand relationships and context between the tokens.

How an LLM Works: Main Steps

4. The model predicts the most likely next tokens based on probabilities, generating the output:



How an LLM Works: Main Steps

5. Detokenization:

The generated tokens are converted back into text:

["Paris is the city of love."]

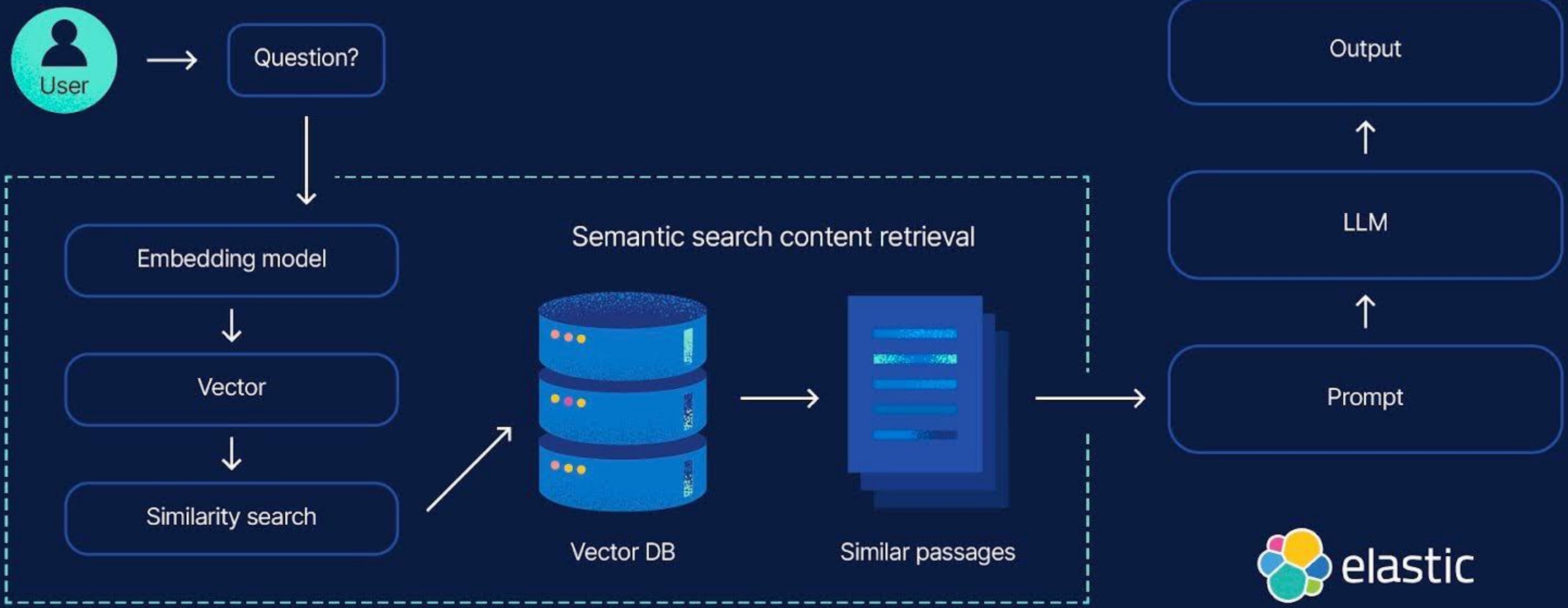
How an LLM Works: Main Steps

6. Result Output:

The final response is presented to the user:

Paris is the city of love.

Retrieval augmented generation (RAG) in action



What is RAG?

> Retrieval-Augmented Generation (RAG) combines:

- **Retrieval**: Fetch relevant data
- **Augmentation**: Enriching the LLM prompt with retrieved data.
- **Generation**: Answer generated by the model

> Why use it?

- Mitigates LLM hallucinations.
- Updates knowledge without training the model.

Tools & Workflow

Key Tools:

Vector DB: FAISS or Chroma (stores document embeddings).

Embedding Model: all-MiniLM-L6-v2 to convert text to vectors.

LLM: Proprietary model (API) or OpenSource (Hugging Face)

Framework (optional): LangChain (quick setup) or custom Python scripts.

ChatGPT

AI Multimodal Agent

Customization

> Main options to consider

- **Data Controls:** Do I want to allow my data to improve the AI model for everyone?
- **Memory:** How can I enable/disable and manage memory settings?
- **Custom Instructions:** How can I instruct ChatGPT about my specific preferences?

[Perplexity AI](#)

AI Search Engine

Customization

> Main options to consider

Preferences:

- Disable “AI data retention”

Personalization:

- Your occupation
- Custom instructions

Prompt

What are real-world problems people face today in [industry or context] that could be solved with a digital product?

Provide:

- 5 concrete problems
- Who experiences them
- Why current solutions are insufficient
- Existing tools (if any)
- Opportunities for improvement or innovation

Prompt

Create a realistic user persona based on real user behavior and complaints related to [context or problem].

Use available data, user feedback, and common frustrations.

Include:

- Name, age, occupation
- Goals and motivations
- Real pain points (based on user complaints or patterns)
- Behaviors and habits
- Tools or apps currently used
- Quote reflecting real user sentiment

Prompt

Based on this user persona, what are the main pain points when trying to solve [problem]?

Include:

- Top frustrations
- What is currently difficult or inefficient
- What the user expects instead

NotebookLM

Think Smarter, Not Harder

Try NotebookLM

Notebook LM
Educational / Multimedia Content Studio

Getting Started

Create a new notebook

Add sources:

- PDFs
- links (articles, websites)
- text or notes

Wait for the content to be processed

Start asking questions about your sources

Prompt

- Summarize the key insights from this report.
- How is AI changing jobs and skills?
- What jobs are most at risk and which are growing?

Prompt Engineering

Fundamentals

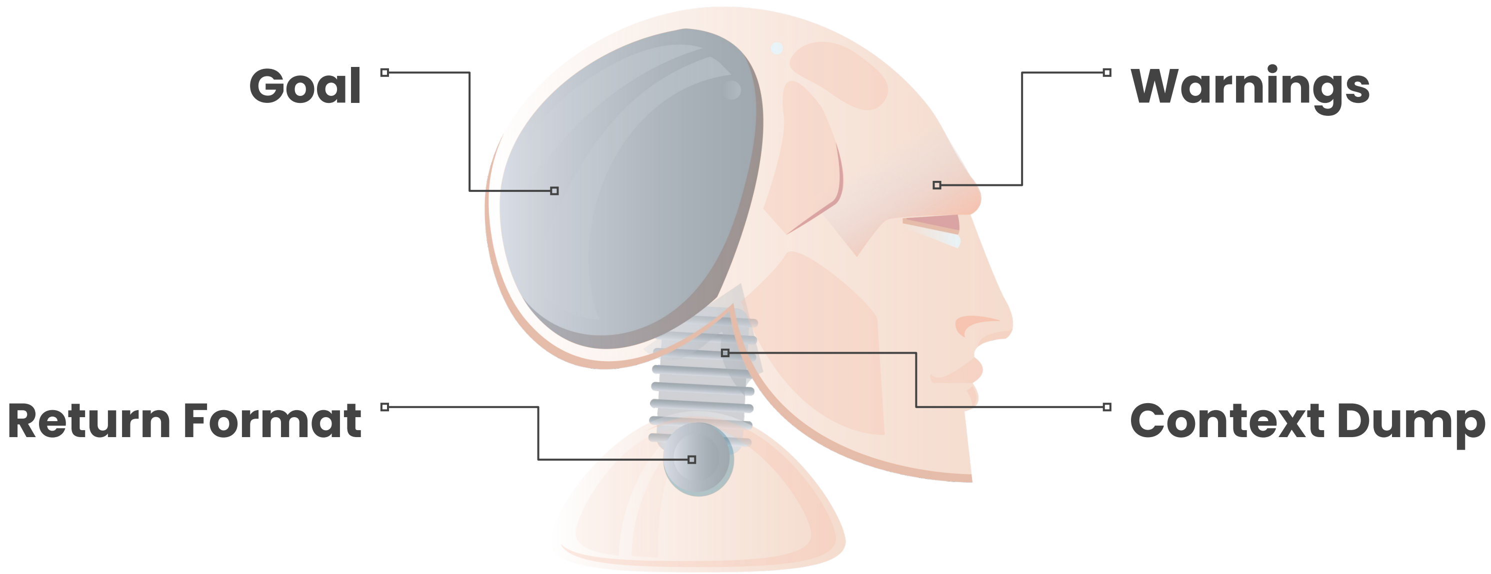
Why is Important

> **Accuracy**: A well-structured prompt can be the difference between a useful answer and an ambiguous or imprecise one.

> **Efficiency**: Reduces the need to repeat or reformulate questions, saving time and resources.

> **Customization**: Allows users to shape and guide the AI's output to meet their specific needs.

Effective Prompt Engineering



Goal

> Define clearly what you expect the AI to accomplish

“Refactor the following legacy JavaScript code to improve readability and modularity.”

“Write a Python function that calculates the optimal charging schedule for electric vehicles based on real-time electricity pricing.”

Return Format

> Specify how you want the AI's output to be structured

“Provide only the refactored code, no explanations.”

“Return a complete function with docstring and type hints.”

“Include comments in the code to explain key changes.”

Warnings

> Clearly indicate potential pitfalls or common mistakes the AI should avoid

“Avoid using external libraries unless absolutely necessary.”

“Do not remove existing validations from the code.”

“Use language-agnostic logic (no platform-specific hacks).”

Context Dump

> Provide essential background information to guide the AI

“This code is part of a Django app for managing EV charging sessions. User authentication is handled via JWT.”

“Include only the frontend logic. Backend API endpoints are already implemented and follow REST conventions.”

“All charging stations are classified as either 'public' or 'private'. This affects the billing logic.”

Avoid

> Generic Prompts (less effective)

"Fix this code."

"Explain this function."

"Build a login form."

"Create a dashboard."

Try Instead

> **Optimized Prompt (more effective and strategic)**

“Refactor this Python script to improve readability and follow PEP8 style guidelines. Keep the logic intact and avoid introducing new dependencies.”

“Explain the following Java method in plain English, focusing on its purpose and how it handles edge cases related to null values.”

Try Instead

> Optimized Prompt (more effective and strategic)

“Build a responsive login form in HTML and Bootstrap that includes client-side validation and links to a forgotten-password flow.”

“Create a frontend dashboard in React that fetches data from a REST API, displays user metrics in charts, and adapts to mobile screens.”

Never CTRL C + CTRL V Blindly

> **Evaluate if your prompt generates useful responses by checking:**

- Does the structure of the output make sense?
- Is the tone aligned with your project?
- Are there concrete examples and added value?

Never CTRL C + CTRL V Blindly

> **If not, consider improving by:**

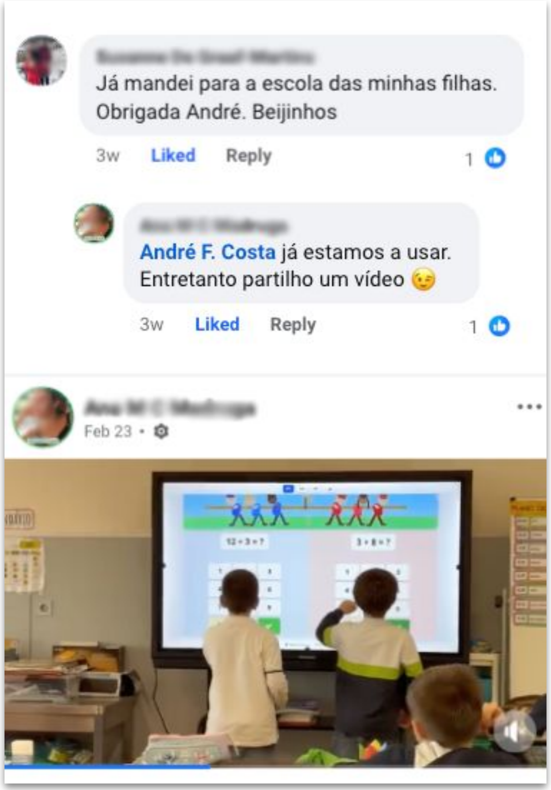
- Clarifying the goal further
- Including additional context or limitations
- Specifying a more suitable response format
- Providing one or more clear examples

AI Generates Ideas

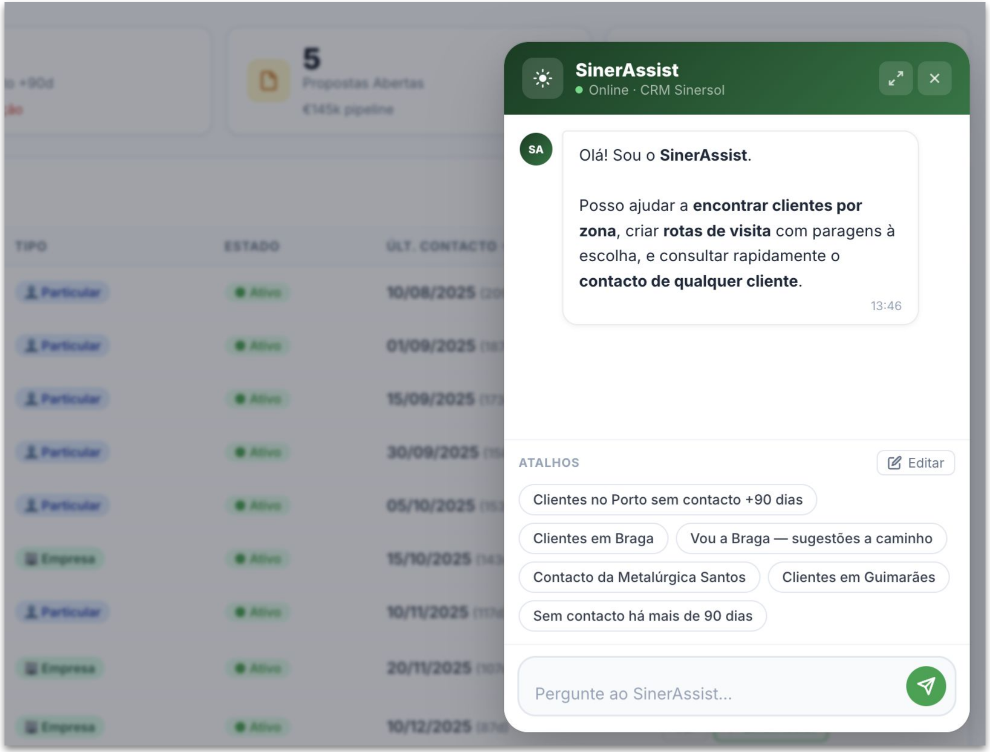
But Users Validate and
Refine The Results

Use Cases

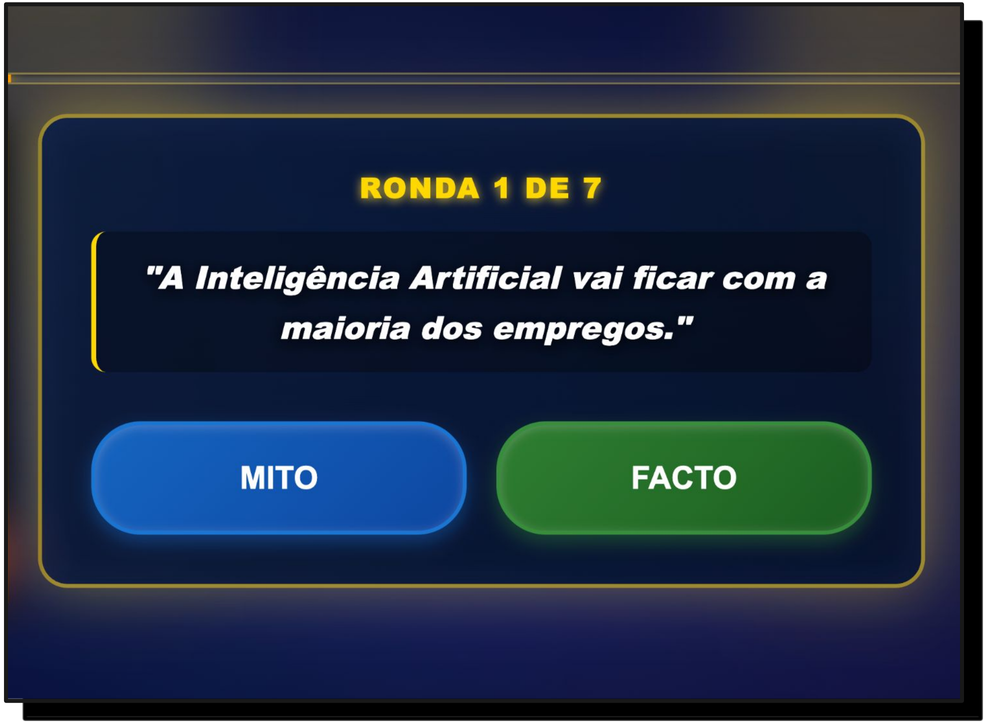
From Idea to Real World



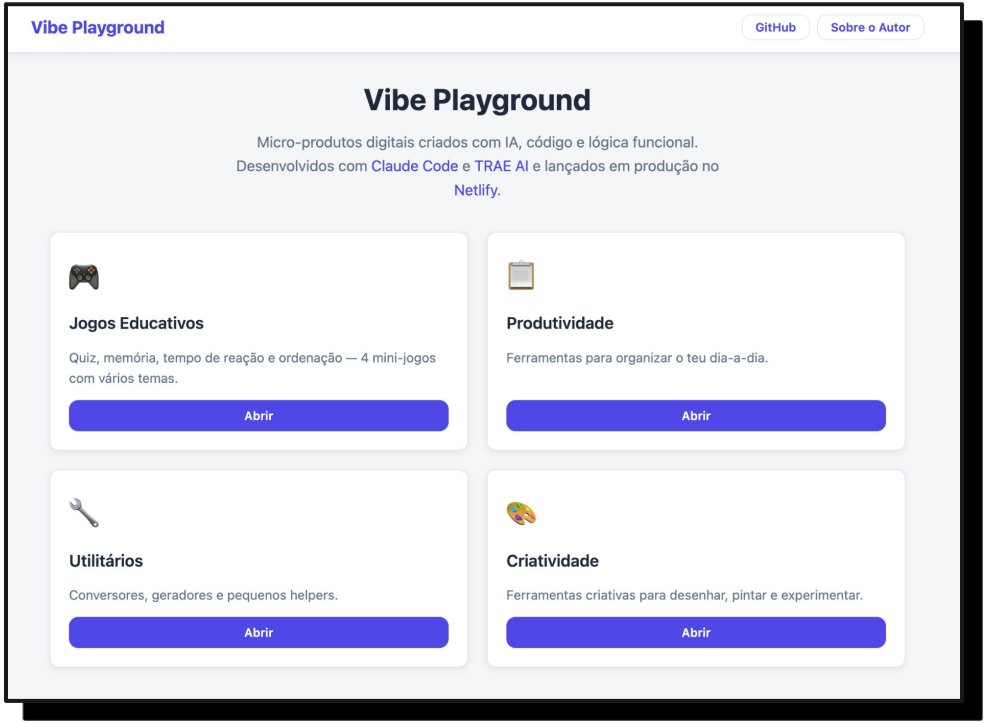
From Idea to Real World



From Idea to Real World



From Idea to Real World

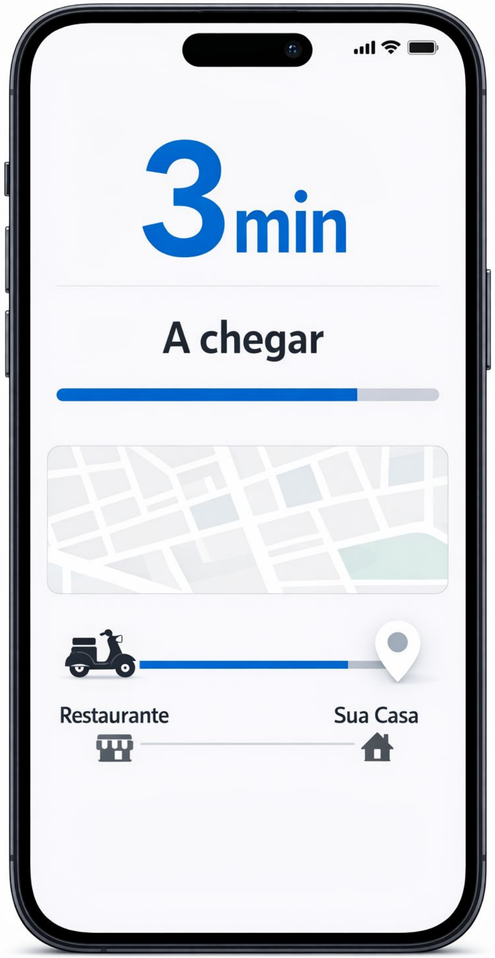


AI in Design

& Digital Product

Sora

Visual Exploration


3 min
A chegar
Restaurante
Sua Casa

3 min
Restaurante
Destino
A Caminho do Destino
Entrega em Andamento
Pedido #1024
1x Combo Promo
Entregador
João Silva
+55 98765-4321
Enviar Mensagem

High-Fidelity Wireframes / Screens

Create a mobile screen (9:16 portrait) for a delivery app that combines map and information in a balanced way to show the delivery status.

The interface should include:

- central map with clearly visible delivery route
- courier's position with a clear marker
- highlighted estimated time of arrival (e.g., "3 min")
- bottom card with delivery status and summary information

Style:

- modern, clean, and functional (avoid excessive colors)
- soft colors with good contrast
- design inspired by real apps like Uber or Glovo
- focus on clarity, not visual effects
- realistic smartphone mockup.

High-Fidelity Wireframes / Screens

Create a mobile screen (9:16 portrait) for a delivery app focused on displaying the arrival time in an extremely clear and immediate way.

The interface should include:

- prominently displayed estimated time (e.g., “3 min”)
- clear status message (e.g., “Arriving”)
- elegant linear progress bar
- simplified location (reduced map or discreet icon)

Style:

- clean, modern, and professional
- neutral colors with emphasis on one main color
- strong typography and clear hierarchy
- organized layout with plenty of white space
- realistic smartphone mockup.


3D Icons

Create a premium 3D icon of a delivery scooter, with a modern and highly detailed style.

The icon should be:

- isolated (no background, transparent background)
- centered in the image
- rendered in a realistic 3D style with a smooth finish
- professional studio lighting
- soft and realistic shadows
- vibrant and contrasting colors (e.g., orange, blue, subtle gradients)
- balanced proportions, modern app icon style

3D Icons

Visual Style:

- Inspired by high-quality 3D icons (Apple, Stripe, Notion)
- Slightly stylized (not entirely realistic)
- Smooth surfaces, no visual noise
- "Premium product design" appearance

Format:

- Square image (1:1)
- High resolution
- No text

43
Equipa
0
Equipa
0
7 + 9 = ?
6 + 6 = ?
PT
EN
FR
37
Equipa 1
0
Equipa 2
0
10 + 3 = ?
8 + 5 = ?
1 2 3
4 5 6
7 8 9
0
1 2 3
4 5 6
7 8 9
0
Sobre o Autor
GitHub

Mockup

Create a photorealistic mockup of a modern classroom in a European school. On the front wall is a large television where the educational game shown in the attached image should appear exactly, fully respecting the layout, colors, and screen elements, without alterations or distortions. The game is interactive (touchscreen).

In front of the TV are two groups of children (four in total, two groups of two), aged between 8 and 11, with ethnic diversity. The children are standing close to the television, touching the screen directly and interacting with the game.

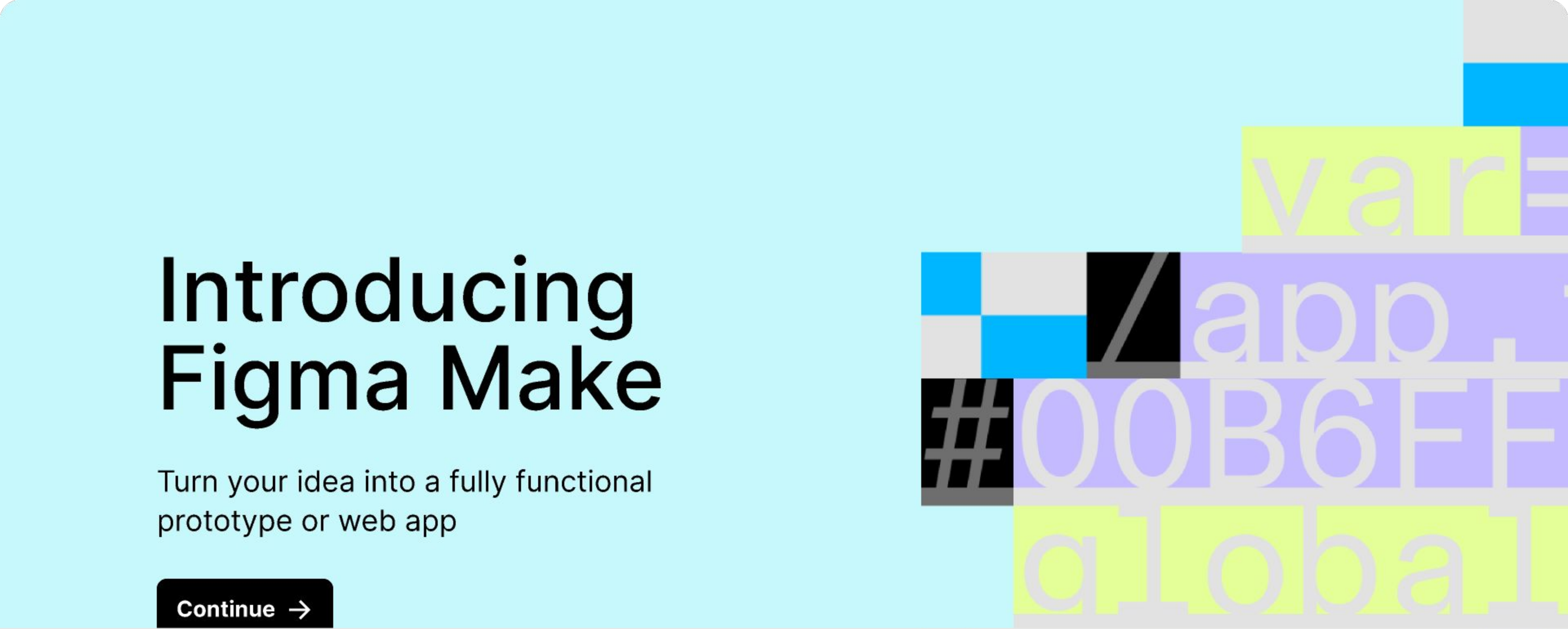
They are engaged, concentrated, and enthusiastic, with active and natural body language. Realistic European school environment, simple and contemporary furniture, natural light entering through the windows. Realistic documentary-style photography, focusing on physical interaction with the screen. No artifacts, no distortions in the game content, no artificial appearance.


Product

Create a realistic studio render of the product based on the provided image, maintaining all proportions, colors, and original identity.

Refine the finish, improve button integration, and subtly develop the top for a more premium look, while maintaining a minimalist and coherent design.

3/4 perspective, soft studio lighting, high resolution, catalog-ready appearance.



[Figma Make](#)

Functional Prototypes

Product

Create a simple interactive prototype for a mobile app that helps users manage daily tasks.

Context:

The app allows users to quickly add tasks and uses AI to automatically organize them by priority.

Task:

Design a clean mobile interface that demonstrates this functionality.

The prototype should include:

- a screen with a task input field

- a list of tasks

- a button called "Prioritize tasks"

Product

- a result state where tasks are automatically reordered

- a small message explaining the change (e.g. "Tasks prioritized based on urgency")

Behavior:

- user adds tasks

- user clicks "Prioritize tasks"

- the list updates with a new order

Design:

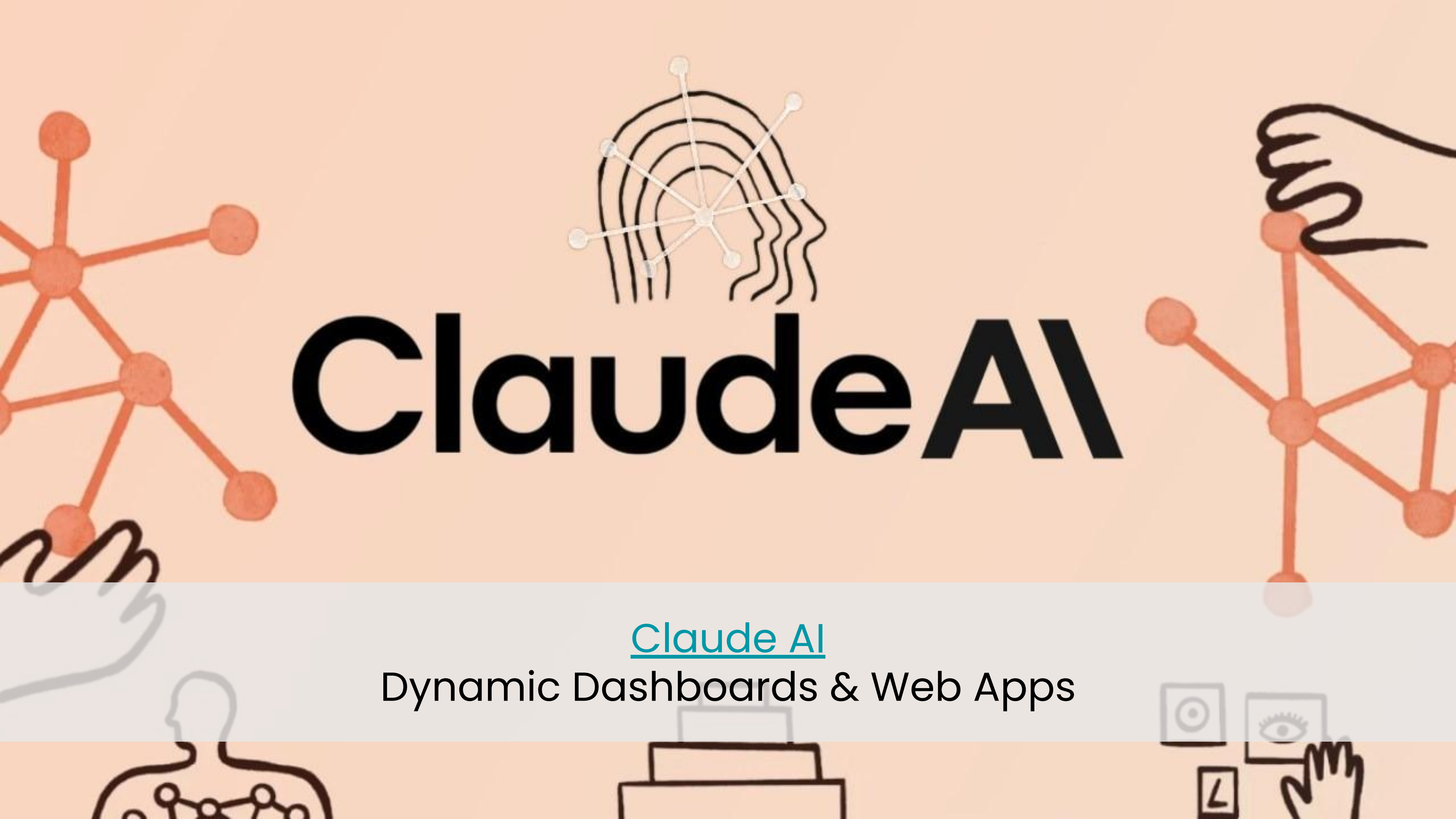
- minimal and clean

- mobile app style

- clear hierarchy

From Vibe Coding

to AI-Assisted Development



Claude AI
Dynamic Dashboards & Web Apps

Product

Create a realistic studio render of the product based on the provided image, maintaining all proportions, colors, and original identity.

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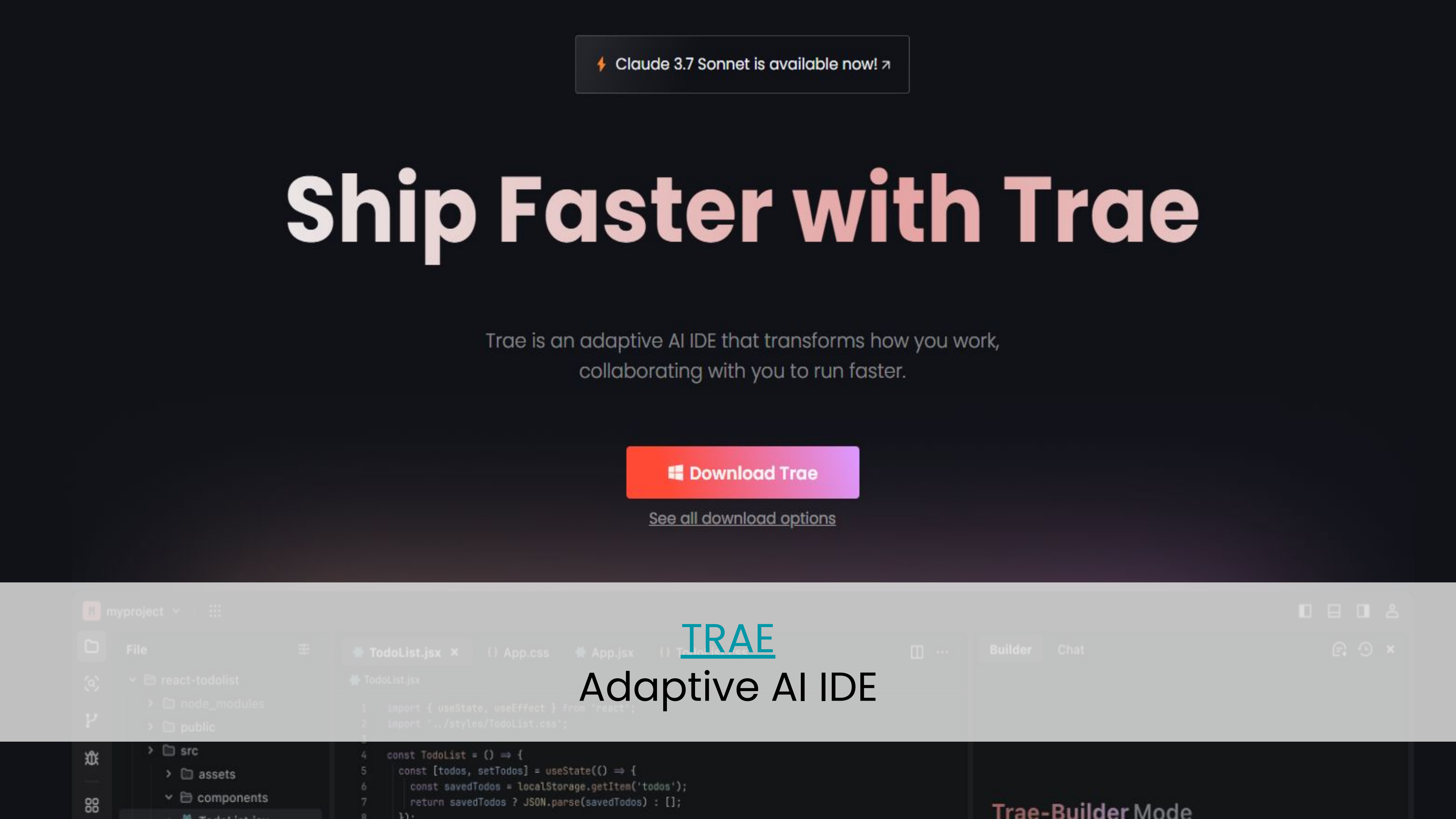

TRAE

Adaptive AI IDE

Product

Create a complete SaaS landing page prototype using only HTML, CSS, and JavaScript.

The fictional company is called BrightFlow.

BrightFlow is a modern SaaS platform for small businesses to manage clients, invoices, tasks, and team collaboration.

The prototype must include:

- a clean and modern hero section

- features section

- pricing section

- FAQ section

- contact/footer section

Product

Add a customer support chatbot with these requirements:

- floating chat button in the bottom-right corner

- when opened, the chatbot should be larger than a typical small widget

- clean, polished SaaS-style UI

- ability to expand the chatbot into a large centered modal

- close button

- message bubbles for user and assistant

- input field and send button

- fake typing/loading effect

The chatbot should work only on the frontend using JavaScript and simulated responses.

Web App Development

& Deploy with AI

Push your ideas to the web

netlify

Netlify

Free Server & Domain Name

Deploy

Publish the Project (Netlify)

- Go to netlify.com
- Create an account / log in
- Click on “Add new site” → “Deploy manually”
- Upload the project folder (HTML, CSS, JS)

The site will automatically go online with a generated link.

Deploy

Change Site Name (Free)

- Go to Site settings
- Select “Change site name”
- Set a custom name

The new link will be: sitename.netlify.app

Note: The free plan allows you to change the site name. Custom domains (e.g., .com) require additional configuration.

Ginni Rometty, former CEO of IBM

"Artificial intelligence will not replace humans, but those who use AI will replace those who don't."

AI Product Studio

Applied Project

Applied Project

From Idea to Web App

Subject: Acting as a digital consulting and development agency, your team's mission is to identify a real-world problem (student life, ecology, local business, etc.) and build a functional Web App prototype to solve it. You will use AI tools (Figma Make, Sora, Claude AI, TRAE AI, or other.) at every stage of the project: ideation, design, and coding.

- Identify a real problem
- Design an AI-supported solution
- Build a functional prototype
- Present your idea and the results

It’s now up to each of us to embrace it actively, experiment with it, and shape the way we work and live.

The future of AI has already started.

That’s a wrap!

> Now it’s your turn — questions, thoughts, curiosities?

Linkedin.com/in/andrecosta/